Groundwater intelligence for Uganda's cattle corridor: integrating monitoring science, open digital technology and drought policy for sustainable dryland water security

Monica Nafula[1], Charles Kakooza[2]

[1] KAI Water & Energy, Kampala, Uganda,

[2]Ministry of Water and Environment

Corresponding author: Monica Nafula, monica.nafula456@gmail.com

Abstract

Uganda's cattle corridor, a belt of semi-arid rangeland running from the south-west to Karamoja, depends on groundwater for people and livestock, yet the evidence needed to manage it is scattered across monitoring files, borehole records and drilling logs. We present an open, integrated digital twin framework leveraging thousands of national borehole records and decades of monitoring records that turns these records into groundwater intelligence and links it to drought and abstraction policy. By interpolating the water levels, depth to bedrock, and yields of 7,306 located boreholes in 2D and 3D and modelling daily levels with transfer-function noise models driven by rainfall and ERA5-Land evaporation, we show that corridor boreholes have deeper water levels than the rest of the country (median 16 m against 10 m). Of 18 corridor stations, 15 show significant trends, 12 of them rising after the wet years of 2019-2024 but declines at Rwonyo in the south-west and at abstraction wells in Mubende show where use, not climate, now controls levels. Station-specific Standardised Groundwater Index thresholds offer operational drought triggers. We set out how these indicators can feed drought bulletins, anticipatory action, and abstraction permits. The open digital twin approach is transferable to other hard-rock drylands.

Keywords: cattle corridor; groundwater drought; Standardised Groundwater Index; digital twin; groundwater governance; Uganda

1 Introduction

Groundwater is the most reliable water source in the world's drylands, where rainfall is short, seasonal, and increasingly erratic (Taylor et al. 2013; Gleeson et al. 2020). In sub-Saharan Africa, groundwater storage has proved resilient to multi-year drought, but this resilience depends on how aquifers are recharged and how heavily they are pumped (MacDonald et al. 2012; Cuthbert et al. 2019). The challenge for sustainability is therefore not only scientific. It requires data that are trusted, technology that turns them into timely information, and policies that act on that information (UNESCO 2022).

Uganda's cattle corridor shows this challenge clearly. The corridor is a belt of semi-arid rangeland that runs from Isingiro and Ntungamo in the south-west, through the central rangelands to Karamoja in the north-east (Fig. 1). It is home to most of the country's livestock and to pastoral communities who depend on boreholes and valley tanks. Rainfall variability and recurrent drought are the main hazards (Nimusiima et al. 2013), and demand is growing for livestock watering and small-scale irrigation.

Uganda already collects much of the evidence needed: a national groundwater monitoring network, a national borehole database and archives of drilling logs. However, these records are rarely analysed together or turned into indicators that drought early-warning systems and regulators can use. This chapter presents an integrated, open framework for groundwater sustainability. It combines (i) monitoring science, using time-series models of groundwater levels; (ii) advanced digital technology, functioning as an open

groundwater digital twin; and (iii) policy, linking indicators to drought response, abstraction control and borehole siting (Fig. 2). We apply it to the cattle corridor and test it nationally to support dryland water security. Section 2 describes the study area and data, Section 3 the methods, Section 4 the results, Section 5 their policy applications and Section 6 the conclusions.

2 The cattle corridor and its groundwater data

The corridor covers 44 of Uganda's 135 districts (2026 boundaries) between about 900 and 1,600 m above sea level. Most of it is underlain by Precambrian granites and gneisses that are deeply weathered to a clay-rich regolith over a fractured transition zone; this weathered-basement aquifer supplies most boreholes (Wright and Burgess 1992; Taylor and Howard 2000). Rainfall is bimodal in the south-west and centre, generally below 1,000 mm per year, and becomes unimodal and lower toward Karamoja. Recharge in the Upper Nile basin is episodic and tied to heavy rainfall (Owor et al. 2009).

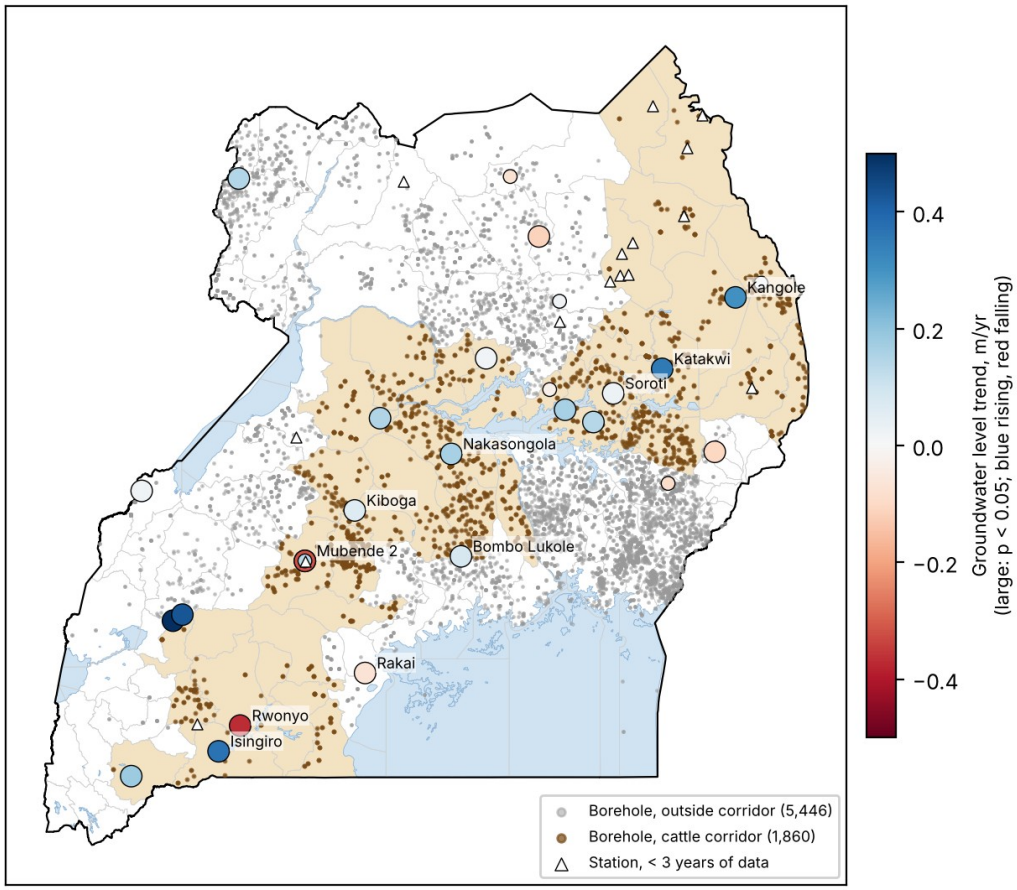


Fig. 1: Uganda's cattle corridor (shaded, 44 districts) with groundwater monitoring stations coloured by observed level trend and the 7,306 located boreholes of the national database. Large circles: significant trend ($p < 0.05$); triangles: stations with less than three years of data. Boundaries: OCHA/UBOS 2026.

Four datasets were used. (1) The national groundwater monitoring network holds daily water levels at 48 stations; 30 lies in the corridor, and 18 of these have at least three years of data, the longest from 1998–2000. Ten telemetric stations installed in Karamoja in 2023 hold about one year of data. (2) The national borehole database holds 12,529 completion records with drilled depth, depth to bedrock, water strikes, static water level, test-pumping yield and water quality. (3) Drilling logs for 5,052 boreholes in the Aswa catchment. (4) Daily precipitation and potential evaporation for 1998–2026 came from ERA5-Land (Muñoz-Sabater et al. 2021).

3 An integrated framework: methods and the digital twin

The framework follows the chain from data to decisions (Fig. 2). To bridge the gap between raw data and actionable policy, the methodology is designed to run as an automated service—an open groundwater digital twin. All steps are open-source Python (Pastas, GemPy, GeoPandas, rasterio), forced by open global data (ERA5-Land), and orchestrated by one command; the repository is versioned on GitHub and archived on Zenodo. Rerun each month as telemetry arrives, this twin keeps the aquifer picture, station models, and drought status current.

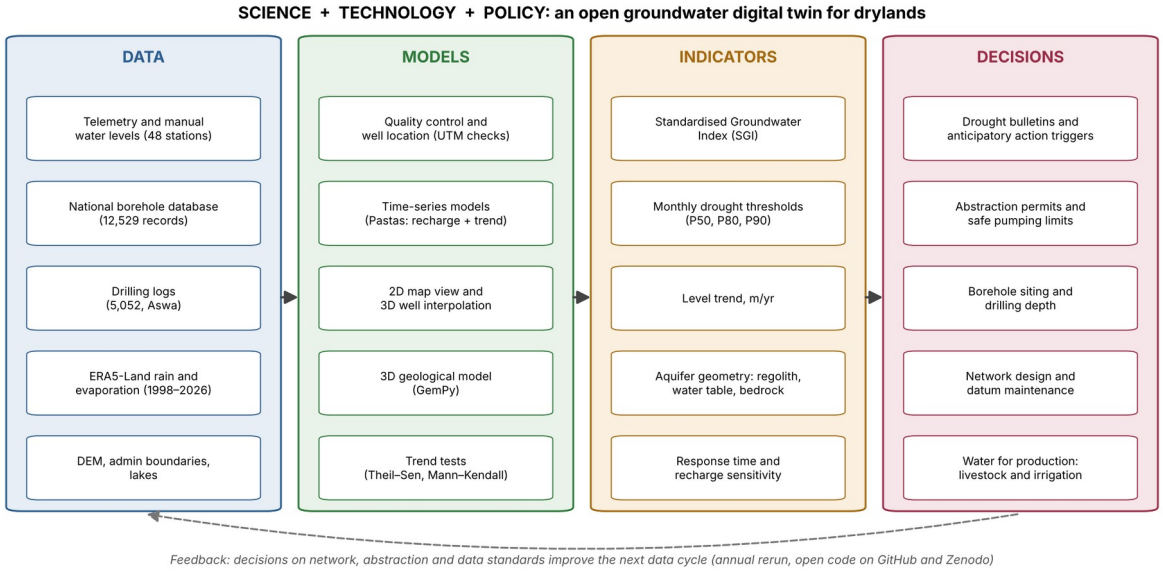


Fig. 2: Framework linking monitoring data, models, indicators and decisions. Feedback from decisions (network design, abstraction records, data standards) improves the next data cycle.

3.1 Data integration and quality control

Borehole coordinates in the national database were tested against multiple hemisphere and zone conventions. Only records falling inside the recorded district were accepted, locating 7,306 boreholes (1,860 in the corridor). Out-of-range values were removed, ground elevation was taken from the DEM, and water-table elevations were computed by subtraction.

3.2 Aquifer geometry in 2D and 3D

Static water level, depth to bedrock, and yield were interpolated in map view by inverse-distance weighting. For the Aswa catchment, drilling logs were classified into cover, weathered or fractured rock, and fresh rock. A 3D implicit geological model was built with GemPy (de la Varga et al. 2019) from 441 boreholes to describe the vertical structure of the aquifer.

3.3 Groundwater level models and drought indicators

Daily groundwater heads were modelled with Pastas (Collenteur et al. 2019; Bakker and Schaars 2019). Recharge was represented as rainfall minus a fitted fraction of potential evaporation, convolved with a Gamma impulse response, and a linear trend was added. ERA5-Land potential evaporation, which averaged about 19 mm/day over Uganda, was rescaled to a long-term mean of 4.5 mm/day. Trends were estimated from monthly mean depths with the Theil–Sen slope (Sen 1968) and tested with the Mann–Kendall test. Groundwater drought was measured by the Standardised Groundwater Index (SGI; Bloomfield and Marchant 2013), with SGI below −1 classed as drought and below −1.5 as severe drought.

For operational use, the 50th, 80th and 90th percentile depths of each calendar month were computed as normal, moderate and severe thresholds.

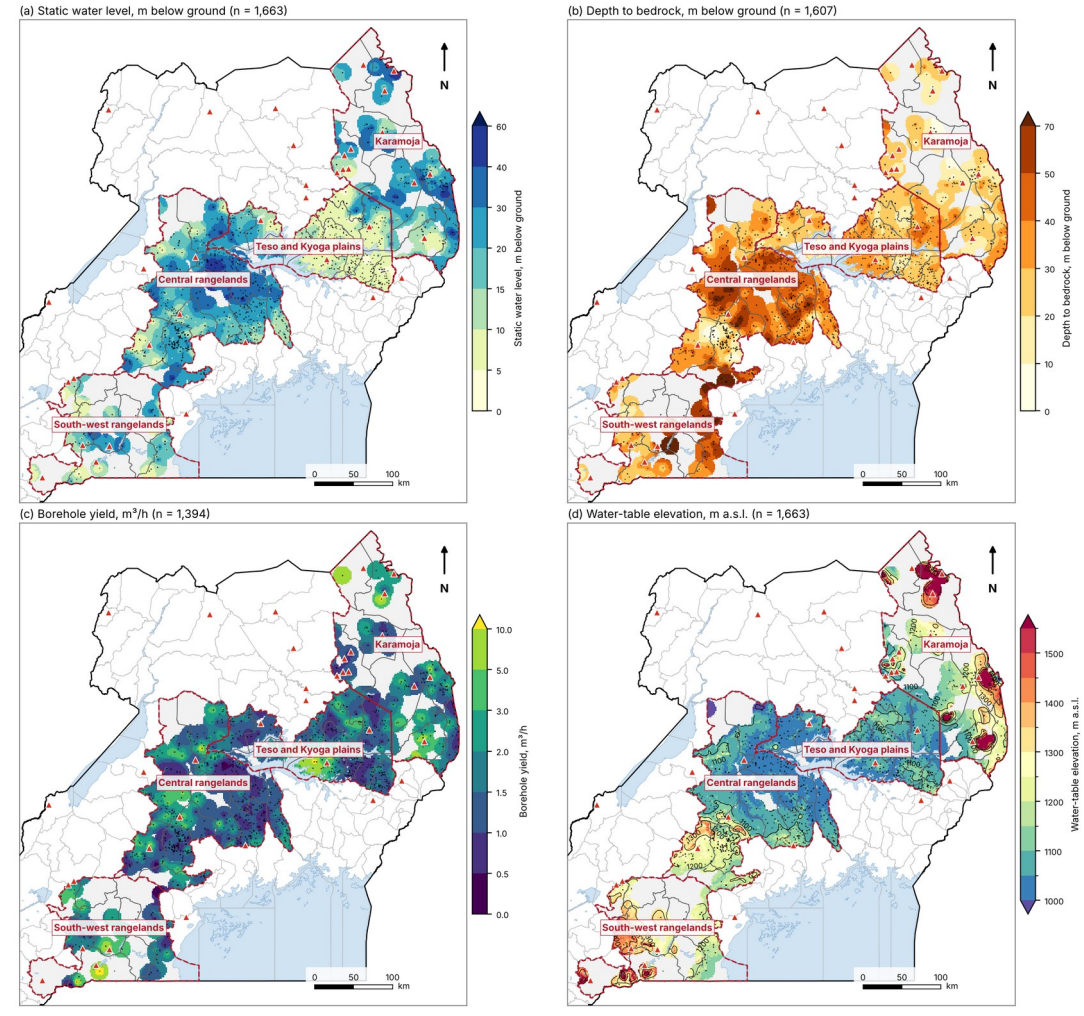
4 Results

4.1 The corridor aquifer in map view and 3D

Corridor boreholes have deeper water than boreholes elsewhere: the median static water level is 16.3 m against 9.7 m elsewhere (Table 1), and 19% of corridor boreholes have water deeper than 30 m against 4% elsewhere. In the central rangelands, where depth to bedrock is about 33 m, the saturated regolith above bedrock is only about 11 m thick (median bedrock depth minus median static water level), against about 17 m in the rest of the country.

Table 1: Borehole characteristics of the cattle corridor sub-regions and the rest of Uganda (medians of located boreholes in the national database; SWL = static water level below ground).

Sub-region	Boreholes	SWL m	Bedrock m	Drilled m	Yield m³/h	Yield < 1 m³/h, %	SWL > 30 m, %
Central rangelands	907	21.6	33.0	70	1.3	23	24
South-west rangelands	136	11.8	31.0	82	2.3	19	10
Teso and Kyoga plains	523	7.5	22.9	64	1.4	28	7
Karamoja	294	20.3	20.0	82	1.5	18	26
Rest of Uganda	5,446	9.7	26.5	61	1.5	22	4



Inverse-distance interpolation (k = 15, power 1.5) with light smoothing; blank where no borehole lies within 12 km. Dots: boreholes; red triangles: monitoring stations. Water table = DEM − interpolated static level. Boundaries: OCHA/UBOS 2026; UTM 36N.

Fig. 3: Cattle corridor in map view: static water level, depth to bedrock, yield and water-table elevation interpolated between boreholes (dots). Areas more than 12 km from a borehole are blank.

The map view shows deep water levels along the central rangelands from Kiboga to Nakasongola, shallow levels across the Teso plains, and patchy deep levels in Karamoja. In 3D, the central corridor shows a continuous regolith aquifer with the water table typically 15 to 35 m below ground.

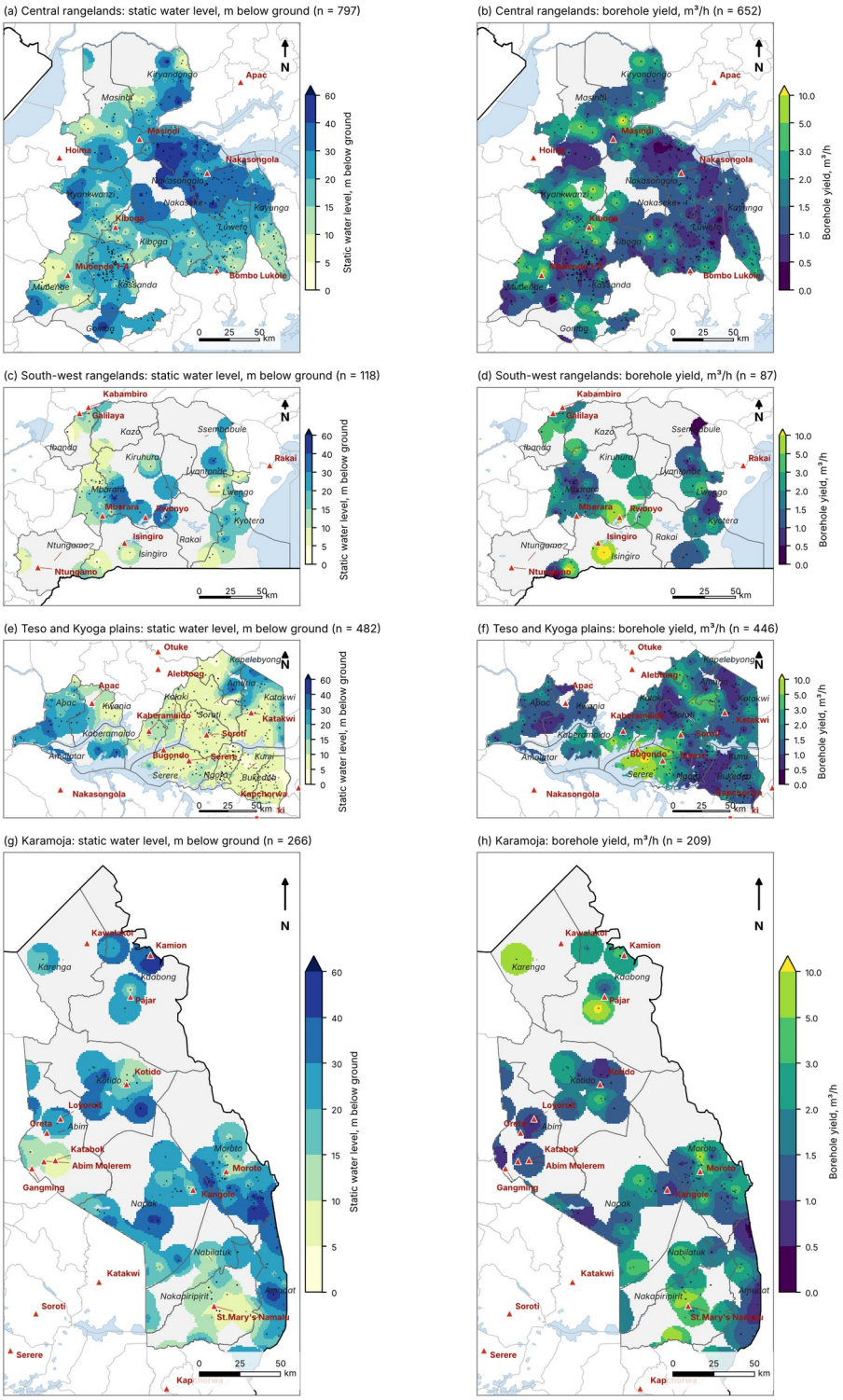


Fig. 4: Zoom into the cattle corridor sub-regions: static water level (left) and borehole yield (right) for the central rangelands, south-west rangelands, Teso and Kyoga plains, and Karamoja, with district names and monitoring stations. Full-page maps of all variables are in the repository (figures/maps).

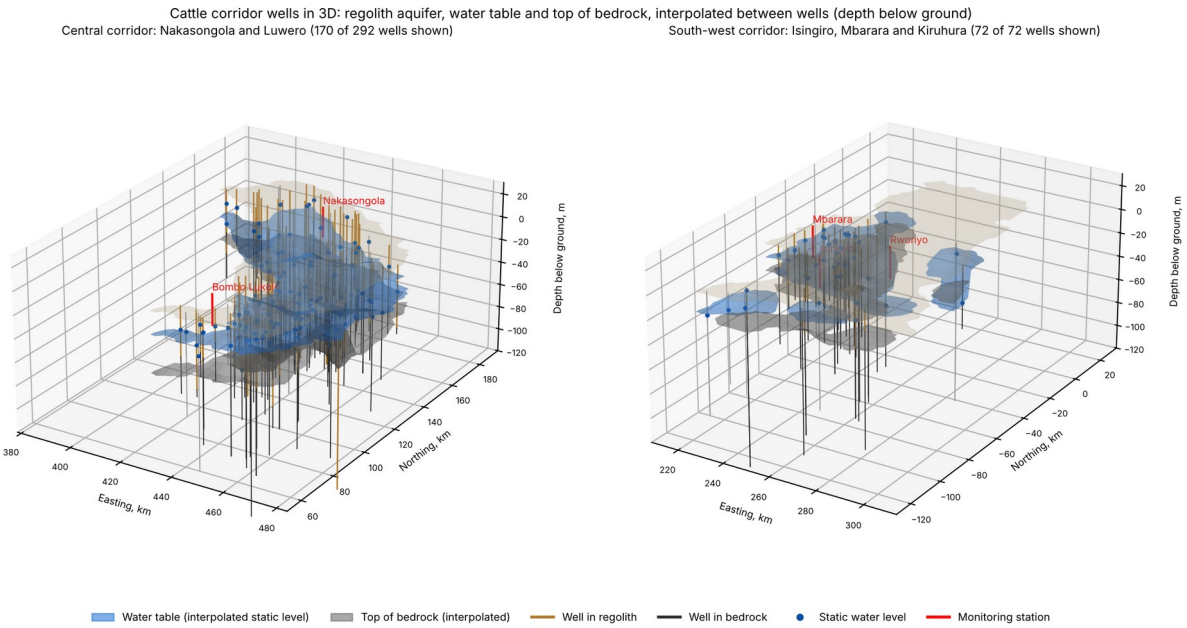


Fig. 5: Wells in 3D for the central corridor (Nakasongola and Luwero) and the south-west corridor (Isingiro, Mbarara and Kiruhura), with interpolated water table (blue) and top of bedrock (grey) in depth below ground. Brown well sections are in regolith, black in bedrock; red bars mark monitoring stations. An interactive version is in the repository.

Nationally, depth to bedrock is about 40 m in the Central region and about 22 m in the Northern region. In the Aswa catchment, drilling logs resolve the vertical profile, showing the productive weathered and fractured zone is about 10 to 12 m thick.

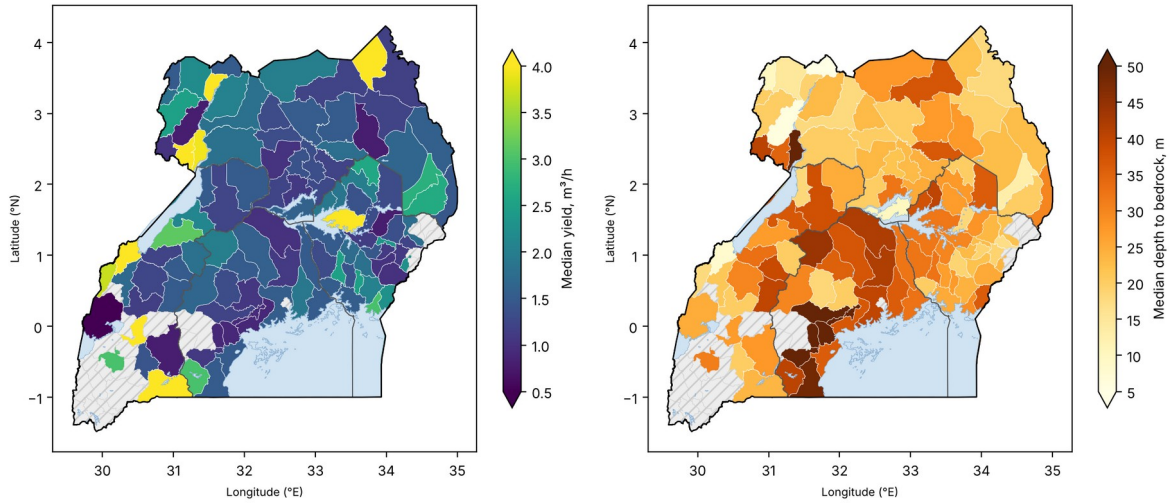


Fig. 6: Median test-pumping yield (left) and depth to bedrock (right) by district from the national borehole database (2026 district boundaries; hatched: fewer than ten records).

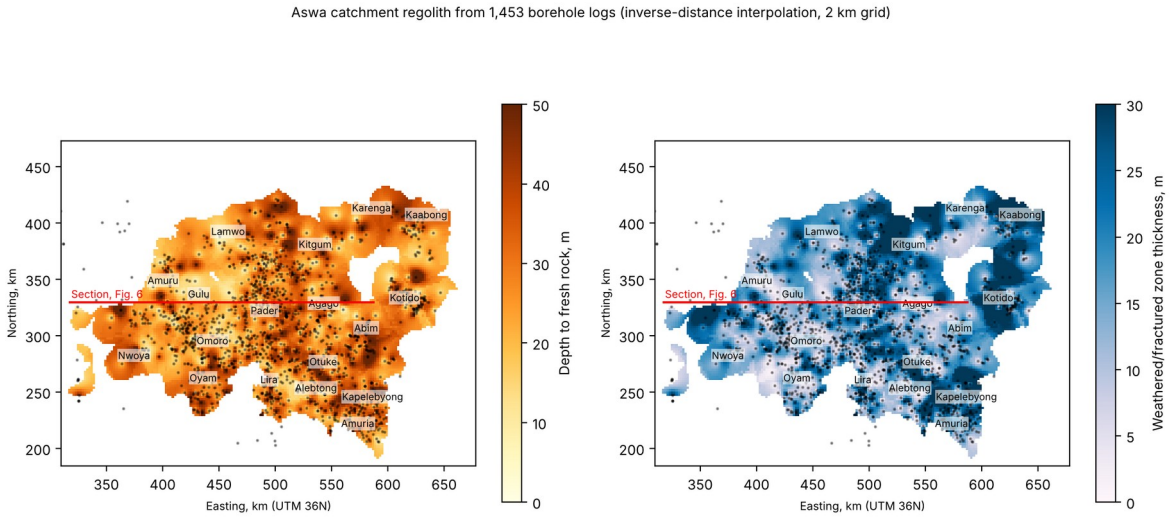


Fig. 7: Aswa catchment: interpolated depth to fresh rock and thickness of the weathered and fractured zone from drilling logs, with district boundaries; the red line marks the section in Fig. 8.

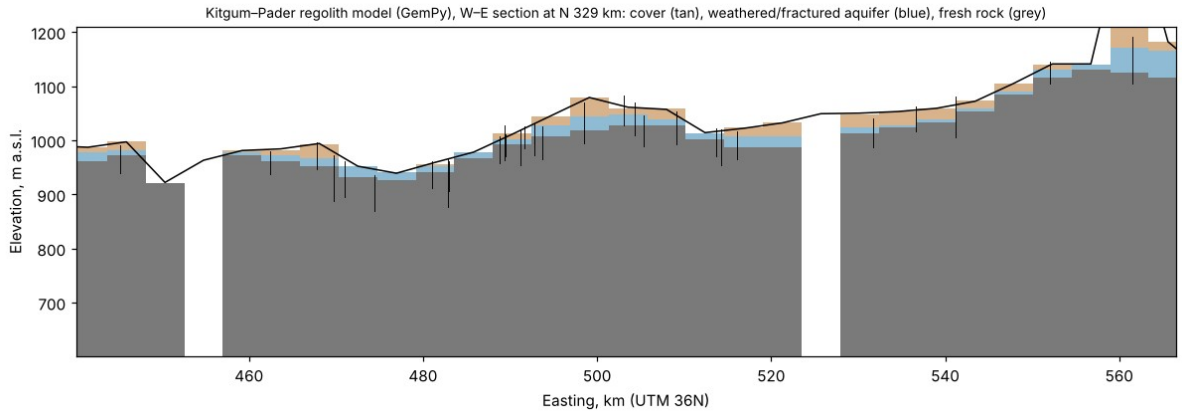


Fig. 8: West–east section through the GemPy model of Kitgum and Pader: cover (tan), weathered and fractured aquifer (blue) and fresh rock (grey), with the DEM surface and boreholes within 3 km. Blank columns have no borehole within 6 km.

4.2 How corridor aquifers respond to climate

With rainfall, evaporation, and a trend, the Pastas models explain at least 60% of level variance at 10 of the 18 corridor stations. Shallow weathered-basement aquifers respond within weeks to months. In contrast, abstraction-monitoring wells (Mubende 1–3, Bombo-Lukole) are not explained by climate alone (explained variance below 10%), which is itself a finding for monitoring policy and abstraction control.

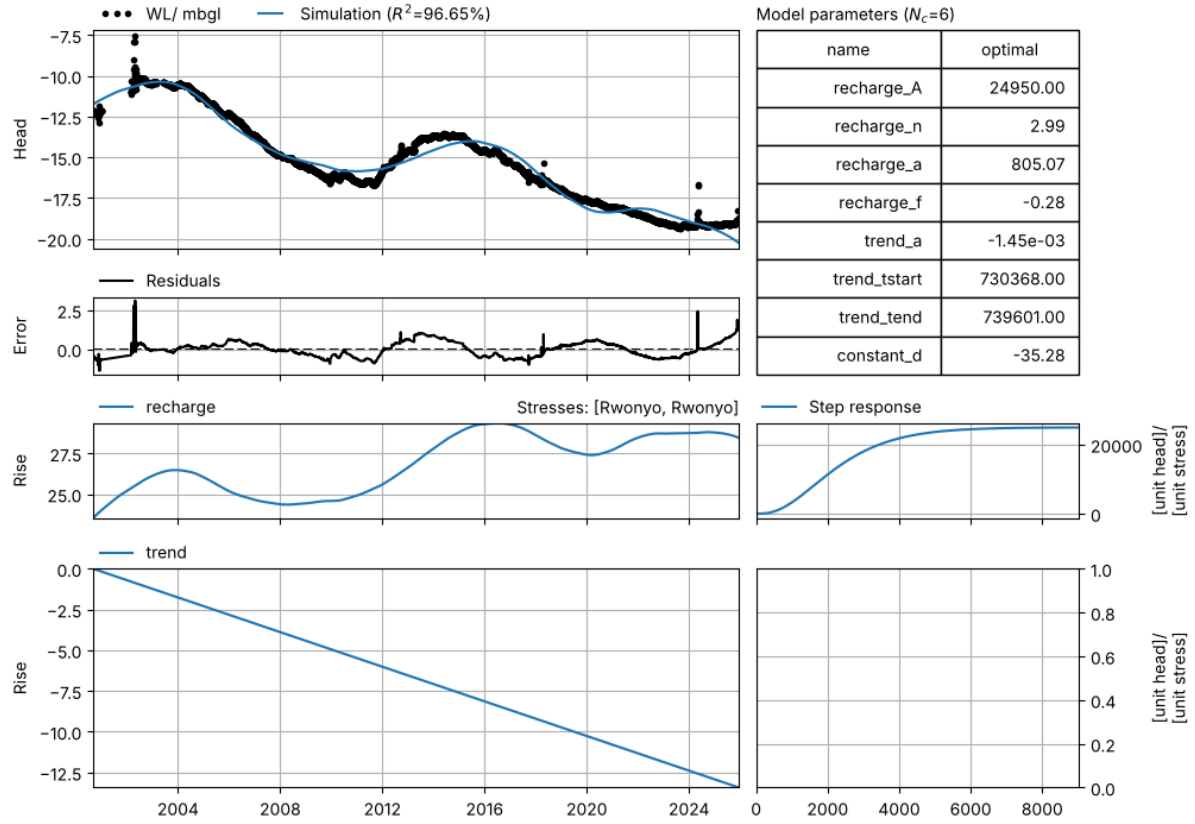


Fig. 9: Pastas model of Rwonyo (2000–2025): observed and simulated head, residuals, recharge and trend contributions, and the recharge step response.

Table 2: Cattle corridor monitoring stations: model fit, response time, observed level trend (Theil–Sen; positive = rising), latest SGI and drought status. EVP = explained variance percentage; half response = time to half of the step response.

Station	District	Record	EVP %	Half response (days)	Trend m/yr	Significant	Latest SGI	Status
Rwonyo	Kiruhura	2000–2025	97	2147	-0.37	yes	-1.08	drought
Masindi	Masindi	2007–2025	90	28668	0.15	yes	1.28	normal or above
Bugondo	Serere	2015–2026	85	1654	0.16	yes	0.47	normal or above
Kangole	Napak	1998–2025	82	981	0.30	yes	0.07	normal or above
Katakwi	Katakwi	2017–2025	81	103	0.35	yes	0.67	normal or above
Nakasongola	Nakasongola	2010–2025	81	52	0.16	yes	1.10	normal or above
Soroti	Soroti	1998–2025	73	3482	0.03	yes	0.45	normal or above
Isingiro	Isingiro	2007–2025	70	712	0.37	yes	1.38	normal or above
Kiboga	Kiboga	2008–2026	68	2447	0.06	yes	1.09	normal or above
Ntungamo	Ntungamo	2007–2026	68	8545	0.18	yes	0.07	normal or above
Apac	Kwania	1998–2026	40	1592	0.02	yes	-0.05	normal or above
Kaberamaido	Kaberamaido	2017–2025	33	126	-0.03	no	1.47	normal or above
Moroto	Moroto	2010–2026	19	318	0.03	no	0.46	normal or above
Serere	Serere	2009–2025	18	46	0.14	yes	0.82	normal or above
Mubende 3	Mubende	2015–2025	7	7	0.14	no	0.52	normal or above
Bombo-Lukole	Luwero	1998–2024	4	7	0.09	yes	-1.96	severe drought
Mubende 2	Mubende	2015–2025	1	3	-0.32	yes	-0.97	below normal
Mubende 1	Mubende	2015–2025	1	4	-0.12	yes	-0.59	below normal

4.3 Trends and groundwater drought

Of the 18 corridor stations, 15 show significant trends (Mann–Kendall $p < 0.05$). Twelve are rising, most strongly at Isingiro and Katakwi, consistent with the exceptionally wet years of 2019–2024. However, three are falling: Rwonyo (−0.37 m/yr) and the Mubende abstraction wells.

Rwonyo's level fell about 9 m between 2002 and 2025. Because the decline continued through the wet years, it points to a local cause, most likely growing abstraction for livestock, rather than climate alone. Bombo-Lukole reached severe drought (SGI −1.96) in 2023–2024.

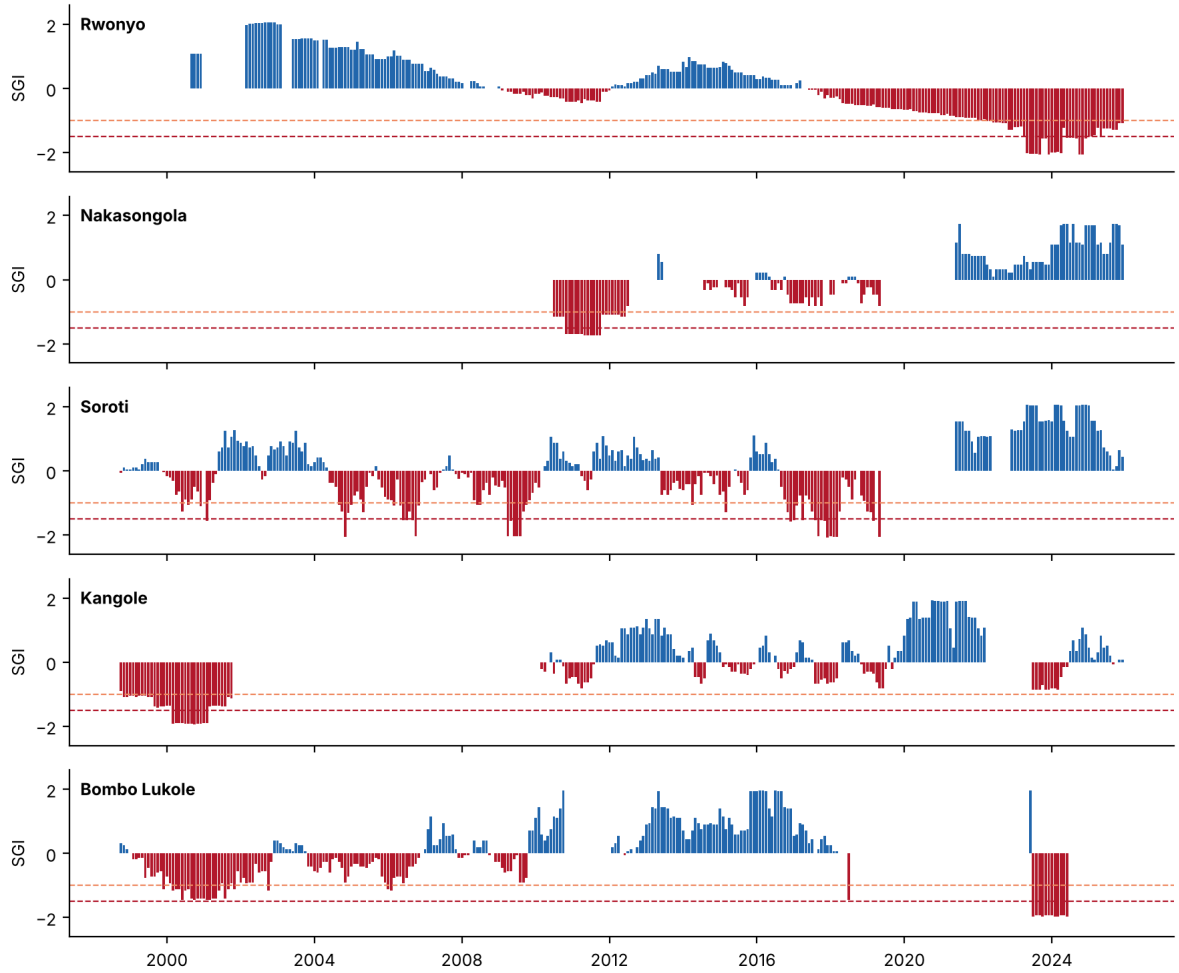


Fig. 10: Monthly Standardised Groundwater Index at five cattle corridor stations. Dashed lines mark drought (−1) and severe drought (−1.5).

Table 3: Example monthly groundwater drought thresholds (depth below ground; m bgl = metres below ground level). A month whose mean depth exceeds the moderate or severe value is in moderate or severe groundwater drought. Thresholds for all stations and months are in the repository.

Station	Month	Normal P50 (m bgl)	Moderate P80 (m bgl)	Severe P90 (m bgl)
Isingiro	Jan	24.69	26.92	28.58
Isingiro	Apr	24.90	28.68	29.12
Isingiro	Jul	24.57	28.42	29.04
Isingiro	Oct	24.65	28.21	28.82
Rwonyo	Jan	15.68	17.98	18.97
Rwonyo	Apr	15.55	17.98	19.03
Rwonyo	Jul	15.58	18.00	19.00
Rwonyo	Oct	15.46	18.13	19.01
Kangole	Jan	9.11	12.20	30.38
Kangole	Apr	10.36	12.01	30.50
Kangole	Jul	9.12	10.30	30.43
Kangole	Oct	8.09	18.82	30.32
Nakasongola	Jan	7.56	8.00	8.86
Nakasongola	Apr	7.63	8.02	9.06
Nakasongola	Jul	7.44	8.22	8.73
Nakasongola	Oct	7.40	7.70	8.72

5 From indicators to policy

The results translate into three policy applications.

Drought early warning and anticipatory action. The SGI thresholds generated by the digital twin can be published in monthly bulletins and adopted as triggers in district anticipatory action protocols. Furthermore, running the station models forward with seasonal rainfall forecasts can give a groundwater outlook three to six months ahead providing the lead time that anticipatory action needs.

Abstraction control for sustainability. The Water Act requires permits for groundwater extraction. The declines at Rwonyo and Mubende, and the severe drought at Bombo-Lukole, suggest that local abstraction can dominate levels. To ensure long-term groundwater sustainability, permit conditions in the corridor should require metered abstraction and reporting. The 3D aquifer model can be coupled with these records to estimate safe yields for livestock and irrigation.

Siting and design of water points. The maps show where water is deep and regolith thin, supporting optimal siting, drilling depth, and pump choice for rural systems.

Limits of the approach. Only 58% of the 12,529 borehole records could be located reliably, ERA5-Land evaporation required rescaling, and inverse-distance interpolation cannot represent local fractures. Abstraction wells cannot be fully interpreted without accurate pumping records, emphasising the need for robust data governance.

6 Conclusions

Uganda's cattle corridor has deeper water and thinner saturated aquifers than the rest of the country. While levels rose at most stations during the recent wet years, locations with concentrated abstraction, like Rwonyo and Mubende, show falling levels despite the rain. Integrating monitoring science, digital twin technology, and policy turns scattered records into station-specific drought triggers and evidence for sustainable abstraction control. The strategic investments needed are modest: maintained telemetry, metered abstraction, and an open, repeatable workflow. The digital twin framework is transferable to other hard-rock drylands across Africa and beyond.

Acknowledgements

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Competing interests

M. Nafula is a co-founder of KAI Water & Energy, which provides hydrological and groundwater services. The analysis uses only open-source software and the code is freely available. The other author declares no competing interests.

Data and code availability

Code, notebooks, derived tables and figures are available on GitHub (https://github.com/Kaitale/uganda-cattle-corridor-groundwater) and archived with this chapter on Zenodo (https://doi.org/10.5281/zenodo.23725130). Raw monitoring, borehole and drilling-log data belong to the Ministry of Water and Environment, Uganda, and are available on request.

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